隐藏翻译（T）

How DNA's discovery changed the world - and my life

【BBC精选丨诺贝尔得主：DNA的发现改变了世界，也让我找到了亲生父亲】

Summary: DNA is the fundamental genetic material in all known life, containing vast amounts of information that could theoretically store all digital data on Earth. Nobel laureate Paul Nurse explains how DNA's double helix structure, discovered in 1953, enables genetic inheritance and identity verification through DNA fingerprints. The sequencing of the human genome in 2000 revolutionized fields from criminal justice to ancestry tracing, while CRISPR gene editing now offers potential cures for genetic diseases. DNA analysis continues to reveal profound insights into human evolution, health, and even personal family mysteries.

摘要： DNA是所有已知生命的核心遗传物质，其信息存储能力足以容纳全球数字数据。诺贝尔奖得主保罗·纳斯阐释了DNA双螺旋结构如何支撑遗传机制，并通过DNA指纹技术实现身份认证。人类基因组计划的完成推动了司法、祖源追溯等领域的变革，CRISPR基因编辑技术为遗传病治疗带来新希望。DNA分析不仅揭示人类进化与健康奥秘，甚至能解开个人身世之谜，持续重塑我们对生命的认知。

⏱ Estimated Reading Time: 8 min.

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DNA contains the genetic code found in all known life on our planet.

DNA包含我们这个星球上所有已知生命体中发现的#遗传#密码。

In each of nearly all of your roughly 30 trillion cells, there are 6.4 billion letters of DNA.

在你体内近30万亿个细胞中，每个细胞都含有64亿个DNA字母。

It's powerful stuff.

它是强大的物质。

If the DNA in all of your cells was used to store computer data,

若将你所有细胞中的DNA用于存储计算机数据，

it could hold the equivalent of all the digital data we currently store on Earth.

其容量可相当于目前地球上存储的所有数字数据。

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I'm Paul Nurse and I spent much of my working life thinking about DNA.

我是保罗·纳斯，职业生涯中大部分时间都在研究DNA。

In particular, how it's copied and distributed inside cells every time they divide.

尤其专注于细胞每次分裂时DNA的复制与分配机制。

I was awarded the Nobel Prize for this work in 2001.

我因这项研究于2001年获得诺贝尔奖。

Our understanding of deoxyribonucleic acid or DNA has grown enormously since its discovery in the 19th century.

自19世纪被发现以来，我们对脱氧核糖核酸（DNA）的理解已取得巨大进展。

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DNA was shown to be responsible for genetic inheritance in 1944.

1944年，DNA被证实负责#遗传# #传递#。

Then in 1953, its structure was revealed using x-rays.

随后在1953年，通过X射线技术揭示了其结构。

And DNA turns out to be a stunningly elegant molecule.

事实证明DNA是一种极其精妙的#分子#。

What you're looking at is the original DNA model currently on display in the Science Museum in London.

您看到的是现陈列于伦敦科学博物馆的原始DNA模型。

It shows two long chains of molecules spiraling around each other in a double helix,

它显示两条分子长链以双#螺旋#结构相互缠绕，

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like a twisted ladder.

如同扭曲的#梯子#。

The rungs are pairs of four chemicals marked as AGCNT.

梯级由标记为AGCT的四种化学物质两两配对构成。

Determining the order of these chemicals, their sequence is known as sequencing.

确定这些化学物质的顺序即其#序列#的过程被称为测序。

Being able to describe the DNA sequence allows scientists to identify important differences between individuals.

描述DNA#序列#的能力使科学家能识别个体间的重大差异。

These unique differences have become the basis for what is known as our DNA fingerprint.

这些独特差异构成了所谓DNA指纹的基础。

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The first ever recorded DNA fingerprint was made in 1984 by my friend Alex Jeffries at the University of Leicester.

史上首个DNA指纹记录由我朋友亚历克·杰弗里斯于1984年在莱斯特大学完成。

Using DNA taken from his technician, Vicki Wilson,

通过分析其#技术员#维姬·威尔逊的DNA，

he described not only which parts of her DNA came from her mother and which from her father,

他不仅解析出她DNA中来自母亲和父亲的部分，

but also the unique genetic code she possessed, one shared by no other human being.

还揭示了她所#拥有#的独特#遗传#密码——这种密码与其他任何人都不相同。

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DNA fingerprints can prove identity, how we are related and more.

DNA指纹可证明身份、亲缘关系等。

DNA testing of hair, skin cells or blood found at crime scenes

对犯罪现场发现的头发、皮肤细胞或血液进行DNA检测，

is now the gold standard for conviction or exoneration of suspects.

现已成为#定罪#或排除嫌疑人的黄金标准。

It has revolutionized the criminal justice system.

它彻底改变了刑事司法系统。

In 2000, the first draft human genome, all the DNA needed to build a human being, was unveiled.

2000年，首个人类#基因组#草图——构建一个人所需的所有DNA——正式公布。

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And with the ability to read genomes came new insights

随着基因组解读能力的提升，

into how the human body works and how we evolve.

我们对人体运作方式及#进化#过程有了新认知。

Analyzing DNA sequences has now reached the public in the form of commercial genetic testing by companies such as Ancestry and 23M.

DNA序列分析现已通过Ancestry和23M等公司的商业遗传检测服务走向大众。

By sending a saliva sample and a fee of course, anyone can receive a report containing information such as where their ancestors were from and who they are related to.

只需寄送唾液样本并支付费用，任何人均可获得包含祖源信息与亲属关系的报告。

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This ability to analyse DNA has had a personal consequence for me.

DNA分析能力也对我个人产生了影响。

It's a twist of fate that although I have a long career in genetics,

命运的#转折#在于：尽管我长期从事#遗传学#研究，

I never realised that my own family had a DNA secret.

却从未意识到自己的家庭隐藏着DNA秘密。

When I was in my 50s, I found out that the person I thought was my sister, was in fact my mother,

五十多岁时，我发现原以为是姐姐的人其实是我母亲，

my parents, the people who raised me were in fact my grandparents.

而抚养我长大的父母实为外祖父母。

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For a long time, the identity of my father remained a mystery,

很长一段时间里，我生父的身份成谜，

but amazingly, now in my 70s, recently it was possible to identify him through DNA testing.

但令人惊叹的是，如今七十多岁的我最近通过DNA检测成功确认了他的身份。

Analyzing DNA has also enabled scientists to do many extraordinary things,

DNA分析还使科学家能完成诸多非凡之事——

ranging from predicting genetic diseases to studying extinct members of the human family

从预测遗传疾病到研究尼安德特人等人类家族灭绝成员，

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like Neanderthals and extinct animals like mammoths.

以及猛犸象等灭绝动物。

A new frontier in genetics is CRISPR,

遗传学的新前沿是CRISPR——

a gene editing tool that works like molecular scissors,

这种基因编辑工具如同分子剪刀，

enabling scientists to cut and paste fragments of DNA within cells.

使科学家能在细胞内剪切和粘贴DNA片段。

This means genetic diseases such as hemophilia, muscular dystrophy and cancers,

这意味着血友病、肌肉萎缩症和癌症等遗传疾病，

in principle might be corrected by editing the DNA of human embryos.

原则上可通过编辑人类胚胎DNA来矫正。

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This has the potential to improve many people's lives, though more research is still required because like all new therapies, they have to be shown to be safe.

虽然仍需更多研究（因为所有新疗法都需证明其安全性），但这有望改善无数人的生活。

There are also questions that need to be answered about making changes that can be passed on to future generations.

那些可遗传给后代的基因修改也引发亟待解答的伦理问题。

But societies have navigated the challenges that come with new scientific technologies before.

但人类社会此前已成功应对过新技术带来的挑战。

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DNA is the stuff of life.

DNA是生命的物质载体。

It is inevitable that the more we research, the more we will understand about ourselves and the living world and the more power we will have to change it.

我们研究越深入，就越能理解自身与生命世界，也越有能力改变它——这是必然趋势。

All this work with DNA will give insights into human development,

所有DNA研究将揭示人类发展奥秘，

allow us to study extinct species and help doctors to treat diseases like cancer with medicine better personalized for everyone's unique genetic makeup.

助力研究#灭绝#物种，并通过更适合每个人独特#遗传# #构成#的药物协助医生治疗癌症等疾病。

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And we are also starting to make new genetic materials

我们已开始制造新型#遗传#材料，

creating new synthetic proteins that could be used for example to help break down pollutants or to reduce greenhouse gases.

创建能分解污染物或减少温室气体的合成蛋白质。

Seven decades after the structure of DNA was first revealed,

DNA结构首次揭示七十年后，

the DNA revolution continues to be exciting.

DNA革命依然激动人心。

It shows no signs of slowing down.

它毫无衰减之势。

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